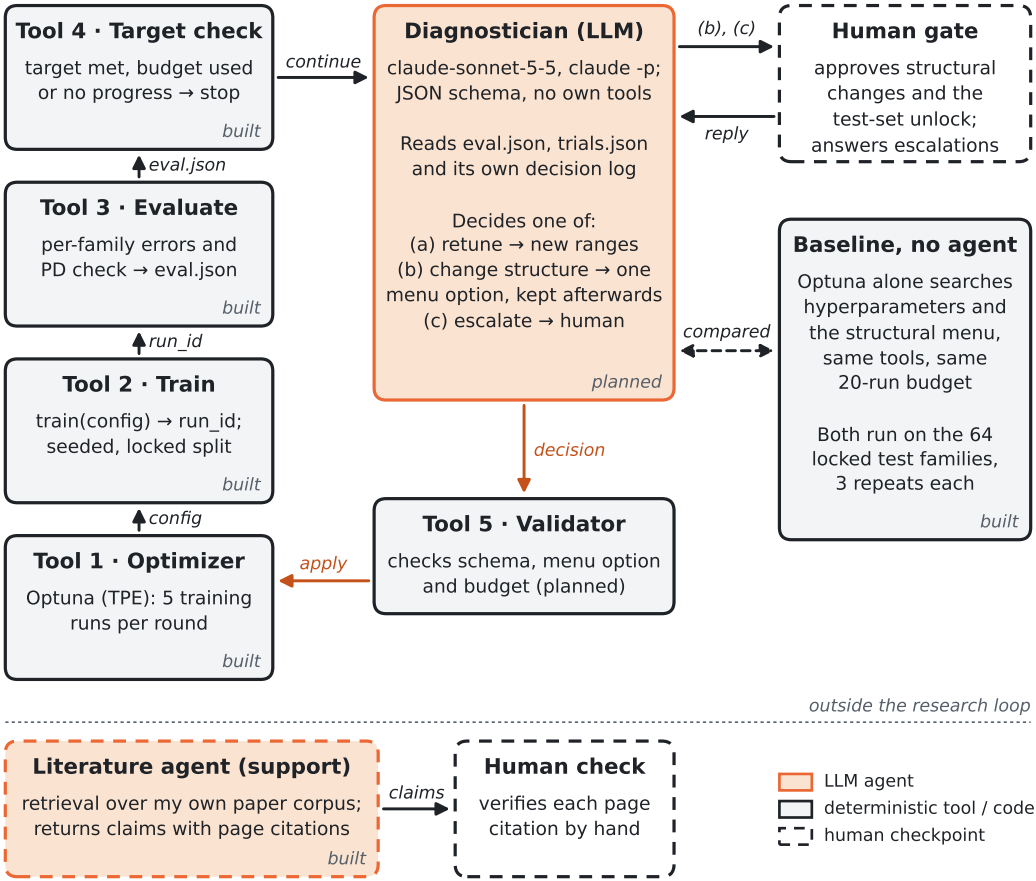
Tool 4 · Target check
target met, budget used or no progress → stop
built
continue
Diagnostician (LLM)
claude-sonnet-5-5, claude -p; JSON schema, no own tools
Reads eval.json, trials.json and its own decision log
Decides one of:
(a) retune → new ranges
(b) change structure → one menu option, kept afterwards
(c) escalate → human
planned
(b), (c)
reply
Human gate
approves structural changes and the test-set unlock; answers escalations
eval.json
Tool 3 · Evaluate
per-family errors and PD check → eval.json
built
run_id
Tool 2 · Train
train(config) → run_id; seeded, locked split
built
config
Tool 1 · Optimizer
Optuna (TPE): 5 training runs per round
built
apply
Tool 5 · Validator
checks schema, menu option and budget (planned)
decision
compared
Baseline, no agent
Optuna alone searches hyperparameters and the structural menu, same tools, same 20-run budget
Both run on the 64 locked test families, 3 repeats each
built
outside the research loop
Literature agent (support)
retrieval over my own paper corpus; returns claims with page citations
built
claims
Human check
verifies each page citation by hand
LLM agent
deterministic tool / code
human checkpoint